

TDMA-CBBA: Communication-Constrained Task Allocation for Drone Swarms with Provable Convergence Guarantees

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Abstract—Decentralized task allocation is a cornerstone of autonomous drone swarm operations, and the Consensus-Based Bundle Algorithm (CBBA) remains its most widely deployed solution, offering convergence guarantees and a 50% optimality bound under idealized, instantaneous, all-to-all communication. Real swarms, however, share a contention-limited radio channel: uncoordinated transmissions collide, packets are lost, and information propagates with bounded but nonzero delay. We present TDMA-CBBA, a communication-constrained extension of CBBA in which agents transmit exclusively within Time Division Multiple Access (TDMA) slots, subject to stochastic packet loss, and maintain time-stamped belief ledgers that expire after a validity horizon T_{val} . Our central theoretical result identifies the predicted operating regime for guaranteed convergence under this model: CBBA’s consensus dynamics are preserved under TDMA when one full TDMA frame fits inside the belief-validity horizon ($N\tau \leq T_{\text{val}}$). Within this regime we prove that (i) consensus terminates within $T_{\text{TDMA}} \leq T_{\text{ideal}} + N\tau K$ for K auction rounds, and (ii) the 50% optimality guarantee of CBBA is preserved despite stale bids. Outside this regime, we show empirically that expired beliefs trigger cascading allocation erasures that provably prevent consensus—a previously unreported *synchronization–staleness phase transition*. We validate our analysis on an open-source benchmark of 1,620 simulation runs across 324 configurations (three slot durations, four baseline radio models) with a full 4,860-run corpus of 15 independent seeds per configuration, spanning swarm sizes of 10–50 drones and packet loss rates up to 30%, quantifying trade-offs between mean time to consensus (31.5 s, 95% CI [30.9, 32.1], right-censored at 50 s), coverage efficiency (0.77), and per-link packet delivery ratio (0.94 at mixed loss). Code, data, and analysis scripts are publicly released.¹

I. INTRODUCTION

Autonomous drone swarms promise transformative capabilities in search and rescue, environmental monitoring, infrastructure inspection, and defense [4], [5]. In all of these missions, a central computational problem recurs: *which robot should do which task?* Centralized assignment is brittle—it introduces a single point of failure, a communication bottleneck at the leader, and a scalability ceiling that grows with fleet size [2], [3]. A standard decentralized answer is the **Consensus-Based Bundle Algorithm** (CBBA) of Choi, Brunet, and How [1],

a consensus auction that provably converges to a conflict-free, 50%-optimal assignment for scoring functions satisfying the Diminishing Marginal Gains (DMG) property, under the paper’s stated communication assumptions. CBBA and its descendants are widely adopted across multi-robot research prototypes [7], [8].

A. Motivation: The Communication Gap

CBBA’s guarantees, however, rest on an implicit physical assumption: that agents exchange their bid ledgers *perfectly, instantaneously, and simultaneously*. No real swarm satisfies this assumption. A fleet of N drones sharing one half-duplex radio channel cannot transmit concurrently; uncoordinated transmissions collide; packets are dropped by fading and interference; and information about a teammate’s bid arrives only after a nonzero, topology-dependent delay [15], [16]. In contested environments—the very settings where swarms are most valuable—adversarial jamming and range limits widen this gap further.

Closing it naively is dangerous. Serializing CBBA’s message exchanges onto a shared channel introduces *information staleness*: each agent acts on beliefs about teammates that are outdated by up to one full communication frame, and stale beliefs interact destructively with CBBA’s conflict-resolution logic. Worse, when practitioners add the standard engineering safeguard of *belief expiry*—discarding telemetry older than a validity horizon T_{val} , so that lost or departed agents are forgotten [16]—we find that staleness can become fatal: if a full TDMA cycle $N\tau$ exceeds T_{val} , every agent’s belief about every teammate expires before it can be refreshed; cascading allocation erasures then race against re-learning, and consensus becomes *structurally unreachable*. To our knowledge, this failure mode—a *synchronization–staleness phase transition* governed by the ratio $N\tau/T_{\text{val}}$ —has not been identified in the CBBA literature, despite belief-expiry heuristics being common practice in deployed systems.

B. Our Approach

We propose **TDMA-CBBA**, a communication-constrained formulation of CBBA in which: (i) agents transmit exclusively

¹<https://github.com/721189/swarm-robotics>

within assigned Time Division Multiple Access slots of duration τ , eliminating intra-frame collisions by construction; (ii) transmissions are subject to independent stochastic packet loss on both transmit and receive legs; (iii) each agent maintains a time-stamped ledger of peers' bids and allocations, filtered through a belief-validity horizon T_{val} ; and (iv) a dynamic slot-allocation layer (D-TDMA) lets agents request doubled transmission time near threats or objectives, prioritizing safety-critical updates. Rather than treating the radio as an afterthought, we place the interplay between scheduling delay ($N\tau$) and belief staleness (T_{val}) at the center of both the analysis and the experimental design.

C. Contributions

This paper makes four contributions:

- 1) **Convergence theory under scheduled, lossy communication (Thm. ??).** We prove that TDMA-CBBA converges to a conflict-free assignment within $T_{\text{TDMA}} \leq T_{\text{ideal}} + N\tau K$ for K auction rounds, provided one full TDMA frame fits within the belief-validity horizon, $N\tau \leq T_{\text{val}}$ (Lemma ??). To our knowledge this is the first convergence result for CBBA under slotted medium access with explicit belief expiry.
- 2) **Optimality preservation despite stale bids (Thm. ??).** We prove that CBBA's 50% optimality guarantee survives stale-bid conflicts induced by bounded communication delay: monotone ledger merging ensures transient misallocations are repaired by subsequent consensus rounds, so the final assignment retains $\mathcal{U}_{\text{TDMA}} \geq 0.5\mathcal{U}$ for DMG score functions.
- 3) **A synchronization–staleness phase transition (Sec. IV-D, Thm. 3).** We prove that when $N\tau > T_{\text{val}}$, belief expiry renders global consensus unreachable, and we validate this impossibility result empirically: trials beyond the transition exhibit consensus rates collapsing toward zero while sub-transition trials converge reliably. This yields an actionable design rule—scale the validity horizon with the TDMA frame.
- 4) **An open-source benchmark with rigorous statistics.** We release a fully deterministic headless benchmark (324 configurations, 15 independent seeds each; swarm sizes 10–50, packet loss up to 30%, three slot durations, four baseline radio models), formal metric definitions, censoring-aware survival analysis, factorial regression, and publication-quality figures. All code, raw data, and analysis pipelines are public.²

D. Paper Outline

Section II formulates the swarm, task, and communication models. Section III details the TDMA-CBBA methodology, including the D-TDMA extension. Section IV presents the theoretical analysis: convergence under ideal communication, convergence and the phase transition under TDMA constraints, worst-case optimality, and complexity. Section V specifies the

TABLE I
SUMMARY OF NOTATION.

Symbol	Meaning
N, M	Number of drones, number of objectives
$\mathbf{p}_i(t), \mathbf{v}_i(t)$	Position, velocity of drone i at time t
$a_i(t) \in \mathcal{O} \cup \{\emptyset\}$	Task assignment of drone i
$b_i(t) \in [0, B_{\text{max}}]$	Battery level of drone i
$s_i \in \{0, \dots, N-1\}$	TDMA slot index of drone i
τ	TDMA slot duration [s]
$p_{\text{tx}}, p_{\text{rx}}$	Transmit-, receive-leg packet loss probability
c_{ij}	Bid (score) of drone i for objective j
$\mathcal{L}_i(t) = (\mathcal{W}_i, \mathcal{B}_i)$	Ledger i : winner map $\mathcal{W}_i$, bid map $\mathcal{B}_i$
T_{val}	Belief-validity horizon [s]
K	Number of auction/consensus rounds
L	Workspace side length (100m)

experimental protocol. Section VI reports results. Section VII discusses limitations and future work, and Section VIII concludes.

II. PROBLEM FORMULATION

We consider a team of N homogeneous drones required to allocate themselves to a set of geographically distributed objectives in a bounded planar workspace, under a contention-limited radio channel. Table I summarizes the notation.

A. Swarm Model

Let $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ be a set of N autonomous drones operating in a bounded two-dimensional environment $\mathcal{W} \subset \mathbb{R}^2$, a square of side $L = 100$ m. Each drone d_i is characterized at time t by:

- position $\mathbf{p}_i(t) \in \mathcal{W}$ and velocity $\mathbf{v}_i(t) \in \mathbb{R}^2$, integrated at timestep $\Delta t = 0.1$ s with first-order steering dynamics;
- battery level $b_i(t) \in [0, B_{\text{max}}]$; a drone whose battery reaches zero is removed from $\mathcal{D}$ (attrition);
- task assignment $a_i(t) \in \mathcal{O} \cup \{\emptyset\}$;
- a local *ledger* $\mathcal{L}_i(t) = (\mathcal{W}_i(t), \mathcal{B}_i(t))$, where $\mathcal{W}_i : \mathcal{O} \rightarrow \mathcal{D} \cup \{\perp\}$ maps each objective to its believed winner and $\mathcal{B}_i : \mathcal{O} \rightarrow \mathbb{R}_{\geq 0}$ stores the corresponding highest known bid;
- a *mailbox* $\mathcal{M}_i(t) = \{(j, \mathbf{m}, t^{\text{rx}})\}$ of time-stamped messages received from peers.

B. Objective and Threat Model

Let $\mathcal{O} = \{o_1, o_2, \dots, o_M\}$ be a set of M static objectives. Each objective o_j has position $\mathbf{q}_j \in \mathcal{W}$ and value $v_j \in \mathbb{R}^+$. The environment additionally contains k surface-to-air-missile (SAM) threat zones $\mathcal{T} = \{\theta_1, \dots, \theta_k\}$; each threat θ_l is a disc $(\mathbf{x}_l, r_l, s_l)$ centered at $\mathbf{x}_l$ with radius r_l and strength s_l that exerts on drone i a repulsive potential

$$U_{\text{threat}}(\mathbf{p}_i) = s_l \left(\frac{1}{d(\mathbf{p}_i, \mathbf{x}_l)} - \frac{1}{r_l} \right)^2, \quad d(\mathbf{p}_i, \mathbf{x}_l) < r_l, \quad (1)$$

where $d(\cdot, \cdot)$ denotes Euclidean distance. Threats shape motion but do not directly destroy drones in this study; agent loss occurs only through battery attrition.

²<https://github.com/721189/swarm-robotics>

C. Communication Model

Communication is governed by a slotted medium-access discipline with stochastic loss:

(C1) *Slotted access (TDMA)*. Time is divided into slots of duration τ ; each drone d_i owns exactly one slot $s_i \in \{0, 1, \dots, N-1\}$ per frame of length $N\tau$, and may broadcast only during s_i . A frame therefore contains exactly one transmission opportunity per drone.

(C2) *Two-stage packet loss*. A broadcast by d_i is lost entirely with probability p_{tx} ; independently, each intended receiver d_j fails to decode the packet with probability p_{rx} . Delivery on link (i, j) thus succeeds with probability $(1 - p_{tx})(1 - p_{rx})$.

(C3) *Belief validity horizon*. Each received message is stored with its timestamp and is considered valid only while $t - t^{msg} \leq T_{val}$. Expired beliefs are purged from $\mathcal{M}_i$, which in turn removes the corresponding peer from drone i 's *perceived swarm*—the set of teammates it believes to be alive and active.

(C4) *Baselines*. Setting $\tau \rightarrow 0$ (all agents transmit every frame) yields an unslotted *open channel*; further setting $p_{tx} = p_{rx} = 0$ yields the *ideal* channel assumed by canonical CBBA [1]. These degenerate cases define the four radio baselines evaluated in Section V.

D. Task Allocation Problem Statement

The objective is a decentralized, conflict-free assignment $a : \mathcal{D} \rightarrow \mathcal{O} \cup \{\emptyset\}$ that maximizes total utility

$$U(a) = \sum_{i=1}^N c_{ii}(a(i)), \quad (2)$$

where $c_{ij} = \max(0.1, C_0 - d(\mathbf{p}_i, \mathbf{q}_j))$ is drone i 's bid for objective o_j ($C_0 = 100$), subject to (a) each objective receiving at most one winner, (b) convergence to ledger agreement within the mission horizon, and (c) operation over the lossy slotted channel of Section II-C. Conflict-freeness is defined by agreement: the allocation is settled when all ledgers coincide, $\mathcal{L}_1(t) = \dots = \mathcal{L}_N(t)$.

III. METHODOLOGY: TDMA-CBBA

TDMA-CBBA interleaves CBBA's two-phase consensus-auction loop with a slotted radio discipline and belief management. Algorithm 1 summarizes one simulation frame; the components relative to canonical CBBA—slotted medium access, two-stage packet loss, and belief expiry—are detailed in Sections III-C–III-E, followed by the D-TDMA extension in Section III-F.

A. Phase I: Consensus Auction over Ledgers

Each drone executes the CBBA phases locally, restricted to its perceived swarm $\mathcal{P}_i$:

(a) *Ghost-drone pruning*. Before bidding, drone i removes from its ledger every objective whose believed winner $w \notin \mathcal{P}_i \cup \{i\}$: it resets $\mathcal{B}_i(o_j) \leftarrow 0$ and $\mathcal{W}_i(o_j) \leftarrow \perp$, releasing the allocation. This mechanism provides fault tolerance against departed agents, but as Section IV-D shows, it is also the mechanism through which belief expiry can destabilize consensus.

Algorithm 1 TDMA-CBBA frame execution (one tick, Δt)

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1:  $t \leftarrow t + \Delta t$ 
     $\triangleright$  // Phase 1: Radio arbitration
2: if slot discipline active then
3:    $s \leftarrow \text{TDMA Scheduler.currentSpeaker}(t)$ 
4:   if  $d_s$  alive and  $\text{rand}() > p_{tx}$  then
5:     enqueue  $(s, d_s.\text{prepareBroadcast}(t))$ 
6:   end if
7: else
8:   for all alive  $d_i$  with  $\text{rand}() > p_{tx}$  do
9:     enqueue  $(i, d_i.\text{prepareBroadcast}(t))$ 
10:  end for
11: end if
     $\triangleright$  // Phase 2: Reception with per-link loss
12: for all enqueued  $(i, \mathbf{m}_i)$  and alive receivers  $d_j, j \neq i$  do
13:   if  $\text{rand}() > p_{rx}$  then
14:      $d_j.\text{receive}(i, \mathbf{m}_i)$ ; update PDR counters
15:   end if
16: end for
     $\triangleright$  // Phase 3: Belief filtering and agent step
17: for all alive  $d_i$  do
18:    $\mathcal{P}_i \leftarrow \{j : t - t_{ij}^{msg} \leq T_{val}\}$   $\triangleright$  perceived swarm
19:    $d_i.\text{auctionAndMove}(\mathcal{P}_i)$ 
20: end for
21: advance scheduler clock by  $\Delta t$ 

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(b) *Local greedy bid*. Drone i evaluates each objective o_j whose score satisfies $c_{ij} > \mathcal{B}_i(o_j)$ and claims the objective with maximal net gain $c_{ij} - \mathcal{B}_i(o_j)$, updating $a_i, \mathcal{B}_i(o_j) \leftarrow c_{ij}$, and $\mathcal{W}_i(o_j) \leftarrow i$; any previously held task is released when switching.

(c) *Ledger merging on reception*. Upon receiving a broadcast from peer k , drone i performs an element-wise monotone merge: if the incoming bid exceeds its own for some o_j , it adopts both that bid and the sender's winner entry. If the adopted winner displaces i 's own assignment ($a_i = o_j$ but $\mathcal{W}_i(o_j) \neq i$), the outbid drone immediately releases $a_i \leftarrow \emptyset$.

Because merges are monotone in bid value and winner entries travel with bids, this update rule is exactly the max-consensus of CBBA [1]; Lemma 1 formalizes the property driving our convergence proofs.

B. Phase II: Motion Control

Between auctions, drones execute a potential-field controller: attraction toward the assigned objective, repulsion from SAM threat zones via Eq. (1), and Reynolds-style separation from neighbors within 4 m—where neighbors are drawn from the perceived swarm $\mathcal{P}_i$ under TDMA (strictly local information), or from ground truth under the open-channel baselines. This asymmetry is deliberate: it makes motion itself sensitive to communication quality, so that SCI responds to both physical flocking and informational coupling. Because this couples two distinct mechanisms (auction communication and motion-state estimation), we additionally run the allocation-only ablation of Sec. V-F, in which motion is disabled for

all baselines, to isolate the contribution of the communication layer to the measured MTC differences.

C. Slotted Medium Access

Under TDMA, exactly one agent broadcasts per scheduling opportunity, eliminating intra-frame collisions by construction. The scheduler advances round-robin through slots, so each agent transmits once per frame of duration $N\tau$; an agent's maximum *inter-transmission gap* equals $N\tau$. This gap is precisely the quantity that couples to the belief horizon T_{val} in Section IV: whenever the gap exceeds T_{val} , the freshest belief about that peer expires before it can be renewed. In the open-channel baselines every alive agent broadcasts every frame, gaps collapse to one tick, and staleness becomes negligible.

D. Two-Stage Packet Loss Model

We model channel impairment at two physically distinct points: (i) *transmit-leg loss* p_{tx} , representing collisions with external traffic, emitter-side fades, or aborted transmissions; and (ii) *receive-leg loss* p_{rx} , representing per-receiver decode failures due to distance-dependent SNR. In our benchmark $p_{\text{rx}} = p_{\text{tx}}/2$ (the receiver leg benefits from coding gain), giving link success probability $(1-p_{\text{tx}})(1-p_{\text{tx}}/2)$. Both stages are independent across links and frames, and every delivery attempt is counted, enabling exact offline computation of PDR.

E. Belief Expiry (Validity Horizon)

Each mailbox entry carries its receive timestamp; entries older than T_{val} are purged before the auction phase. Belief expiry serves two purposes: it bounds mailbox memory, and it implements graceful degradation under agent attrition (departed agents stop transmitting and are forgotten within T_{val}). Its interaction with TDMA scheduling is the central object of Section IV: expiry converts bounded delay into bounded *belief age*, and consensus survives only while worst-case belief age stays below the horizon.

F. Dynamic TDMA (D-TDMA) Extension

Fixed slot lengths waste capacity on quiet agents and starve safety-critical ones. We therefore extend the scheduler with demand-driven slot doubling:

- 1) Each drone continuously evaluates the proximity predicate

$$\phi_i = \left(\min_l d(\mathbf{p}_i, \mathbf{x}_l) < r_l + 5 \text{ m} \right) \vee \left(\min_j d(\mathbf{p}_i, \mathbf{q}_j) < 20 \text{ m} \right), \quad (3)$$

i.e., the drone is near a threat or approaching an objective.

- 2) If ϕ_i holds, the drone raises a double-slot request flag addressed to the scheduler.
- 3) When the drone's next scheduled slot begins, the scheduler grants a doubled slot of length 2τ , then clears the flag.
- 4) Grants are non-preemptive and preserve round-robin order: at most one doubling occurs per visit, slot order is never permuted, so the frame remains collision-free

and the analysis of Section IV carries over with effective cycle time bounded by $2N\tau$.

D-TDMA thus adds an adaptive, safety-aware layer on top of deterministic medium access: contested agents obtain up to twice the telemetry bandwidth exactly when their state estimates matter most (threat evasion, terminal approach), while Theorems ?? and ?? remain valid under the relaxed condition $2N\tau \leq T_{\text{val}}$.

IV. THEORETICAL ANALYSIS

This section develops the paper's formal guarantees under a communication model that is honest about stochastic loss. Section IV-A defines the *belief-age process*—the exact quantity whose interplay with scheduling produces both the guarantees and the failure mode. Sections IV-B–IV-C then state what does and does not survive: a deterministic convergence guarantee in the noiseless regime, a probabilistic guarantee under stochastic loss, and a structural impossibility beyond the staleness threshold. Section IV-E treats optimality under explicitly frozen scores, and Section IV-F analyzes complexity.

Throughout, $q = (1 - p_{\text{tx}})(1 - p_{\text{rx}})$ denotes the per-frame end-to-end delivery success probability on a link, and all probabilistic statements are with respect to the independent loss process of Sec. II-C.

A. The Belief-Age Process

The central object is not the delay of a single packet but the *age* of each agent's belief about each peer.

Definition 1 (Inter-success gap): Fix a directed link ($j \rightarrow i$). Agent d_j transmits once per TDMA frame (period $N\tau$); each attempt survives the sender leg with probability $u = 1 - p_{\text{tx}}$ and is decoded by d_i with probability $r = 1 - p_{\text{rx}}$, independently across frames and links, giving per-frame success $q = ur$. The *inter-success gap* G_{ij} is the elapsed time between consecutive packets from d_j received by d_i : it takes values $G_{ij} = k N\tau$ ($k \geq 1$) with

$$\Pr[G_{ij} = k N\tau] = (1 - q)^{k-1} q, \quad (4)$$

and expected value $\mathbb{E}[G_{ij}] = N\tau/q$.

A belief about d_j held by d_i expires exactly when the current gap exceeds the validity horizon: $\mathcal{K}[G_{ij} > T_{\text{val}}]$. The geometry of Eq. (4) is captured by a single integer.

Definition 2 (Loss-immunity budget): The *loss-immunity budget* of the configuration $(N, \tau, T_{\text{val}})$ is

$$\Lambda = \left\lceil \frac{T_{\text{val}}}{N\tau} \right\rceil - 1 \geq 0, \quad (5)$$

the maximum number of consecutively lost frames a link can absorb while the belief remains valid up to the moment of refresh. Equivalently, expiry occurs on a link iff its gap spans $\Lambda + 2$ or more frame periods.

Two per-link quantities follow immediately from Definitions 1–2:

$$\underbrace{\pi_{\text{exp}} = (1 - q)^{\Lambda+1}}_{\text{expiry prob. per refresh opportunity}} \quad \underbrace{\frac{N\tau}{q} \leq T_{\text{val}}}_{\text{mean-gap (freshness) condition}}, \quad (6)$$

where the second is equivalent to $q \geq N\tau/T_{\text{val}}$. The budget Λ cleanly unifies the regimes discussed in this paper: $\Lambda = 0$ (i.e., $N\tau \leq T_{\text{val}} < 2N\tau$) means a single lost frame already triggers expiry; larger Λ buys robustness linearly in T_{val} ; and the D-TDMA extension of Sec. III-F, by stretching granted slots to 2τ , at most doubles the effective frame period and therefore *halves* the immunity available near threshold.

This decomposition also separates the two loss mechanisms functionally: p_{tx} enters only through the update rate $u = 1 - p_{\text{tx}}$ (it suppresses whole transmissions, inflating gaps toward the mean condition), whereas p_{rx} acts as pure per-link erasure inside q . Empirically this split is exact in our simulator (Sec. VI): measured per-link PDR equals $r = 1 - p_{\text{rx}}$ conditioned on transmission, and measured successful-update frequency equals u .

Lemma 1 (Monotone ledger merging): For every objective o_j and agent i , the stored bid $\mathcal{B}_i^{(t)}(o_j)$ is non-decreasing in t except when reset by ghost pruning, and merging never decreases either party's bid. Moreover $\mathcal{W}_i(o_j)$ always names the agent holding the bid $\mathcal{B}_i(o_j)$.

Proof: Both the reception merge (adopt only strictly greater bids) and the self-bid update (claim only when $c_{ij} > \mathcal{B}_i(o_j)$) write values strictly greater than the stored value; winner entries are copied together with bids from the same packet, so bid-winner pairs remain consistent. Ghost pruning resets a bid to 0 only when its believed winner has left the perceived swarm—which under expiry-free operation cannot be caused by scheduling alone, only by genuine attrition. $\square$

B. Guarantee I: Noiseless Regime (Deterministic)

We first state what holds exactly, without probabilistic qualification.

Theorem 1 (Noiseless convergence): Assume DMG scores (A1), slotted access with $p_{\text{tx}} = p_{\text{rx}} = 0$ (hence $q = 1$, $G_{ij} = N\tau$ deterministically), and the freshness condition

$$N\tau \leq T_{\text{val}}. \quad (7)$$

Then no belief ever expires, ghost pruning never fires from scheduling alone, and TDMA-CBBA reaches a conflict-free consensus within

$$T_{\text{TDMA}} \leq T_{\text{ideal}} + N\tau K, \quad (8)$$

where T_{ideal} and K are canonical CBBA's convergence time and round count on the same instance.

Proof: With $q = 1$ every gap equals exactly $N\tau$, so under (7) beliefs are refreshed strictly before expiry; perceived swarms always contain all agents and pruning is inert. Each logical CBBA round requires one full frame of serialization, stretching round time from Δt to $N\tau$; monotone merging (Lemma 1) ensures each delivered ledger makes irreversible progress toward the unique fixed point that ideal CBBA attains after K rounds [1]. Summing yields (8); conflict-freeness follows from bid-winner consistency. $\square$

This is the precise, deterministic content of Contribution 1: it replaces the earlier claim of bounded delay under stochastic

loss—which cannot hold since $(1-q)^k > 0$ for every k —with a guarantee that is unconditional exactly when loss is absent.

C. Guarantee II: Stochastic Regime (In Probability)

Under genuine loss ($q < 1$), gaps are unbounded random variables (Eq. (4)); deterministic bounds are impossible, and over an unbounded horizon even rare expiry events occur almost surely. The correct statement is therefore *episode-level, in probability*.

Definition 3 (Consensus episode): An episode is a contiguous interval of $F = \lceil K \rceil$ frames during which the swarm attempts consensus from an initial ledger state. It is *expiry-free* if no link experiences $G_{ij} > T_{\text{val}}$ anywhere in the interval.

Theorem 2 (Probabilistic episode consensus): Assume A1, slotted access, $0 < q < 1$, and $\Lambda \geq 0$ per Eq. (5). Then TDMA-CBBA reaches a conflict-free consensus within $T_{\text{ideal}} + N\tau K$ with probability at least

$$\Pr[\text{consensus in episode}] \geq (1 - \pi_{\text{exp}})^{n(n-1)F}, \quad \pi_{\text{exp}} = (1-q)^{\Lambda+1}, \quad (9)$$

where $n = |\mathcal{D}|$ and F is the episode length in frames (Def. 3). The bound is tight in the sense that a single expiry event on any link invalidates the precondition of Theorem 1 for that episode.

Proof: Condition on the episode being expiry-free. There are $n(n-1)$ directed links, each observing at most F refresh opportunities subject to independent loss; by Eq. (6) a given link stays expiry-free across the episode with probability $(1 - \pi_{\text{exp}})^F$, and links are independent, giving the exponent. Conditional on expiry-freeness every belief is refreshed before purging exactly as in Theorem 1, so the same monotone-progress argument delivers consensus within the same time bound. Tightness follows from Sec. III-A(a): one purged belief removes its peer from the perceived swarm and activates ghost pruning for the objectives it held. $\square$

Equation (9) is the probabilistic reliability boundary promised in the introduction: failure probability scales exponentially better in the immunity budget Λ (i.e., in $T_{\text{val}}/N\tau$), and exponentially worse in loss $(1-q)$, network size n , and episode length F . Because $\Lambda + 1$ is an integer step function of $T_{\text{val}}/N\tau$, crossing the threshold jumps π_{exp} from near zero to order unity—explaining why experiments exhibit a sharp empirical transition rather than soft degradation.

[Permanent consistency under stochastic loss] Definition ?? demands agreement for all $t \geq T_{\text{cons}}$. Under $q < 1$ with independent losses this cannot be guaranteed over an infinite horizon: arbitrarily long failure streaks occur almost surely, eventually triggering expiry and re-entering the dynamics. All stochastic guarantees here are therefore episode-granularity (Def. 3); permanent eventual consistency holds only in the noiseless regime of Theorem 1.

D. Structural Impossibility Beyond the Threshold

Theorem 3 (Impossibility for zero-immunity configurations): If $p_{\text{tx}} = p_{\text{rx}} = 0$ and $N\tau > T_{\text{val}}$ (i.e., $\Lambda < 0$), then for every agent i and peer $j \neq i$ the belief about j expires before j 's next

transmission. Consequently each perceived swarm collapses to $\{i\}$ between receptions, ghost pruning erases all foreign winner entries every auction round, and no allocation survives a full frame: consensus (Def. ??) is unreachable for any $N \geq 2$, deterministically, regardless of runtime. For $q < 1$ with $N\tau > T_{\text{val}}$ the collapse occurs with probability tending to one within $O(1)$ frames.

Proof: With $q = 1$, consecutive transmissions of d_j are spaced exactly $N\tau > T_{\text{val}}$; just before each transmission, every earlier packet from d_j has aged past T_{val} and is purged by (C3). Hence $\mathcal{P}_i$ excludes j whenever d_i auctions, and rule (a) of Sec. III-A resets all winner entries belonging to j ; symmetrically j erases i 's wins. Allocations created in one frame are destroyed in the next, cycling forever. For $q < 1$: a gap $\leq T_{\text{val}} < N\tau$ would require receiving two packets from j within a single inter-transmission spacing—impossible under one-transmission-per-frame scheduling—so perceived swarms collapse identically, now with probability one over any window longer than a few frames. $\square$

Together, Theorems 1–3 locate the operating envelope: the ratio $T_{\text{val}}/N\tau$ determines whether the swarm sits in the deterministic regime ($q=1$, $\Lambda \geq 0$), the probabilistic regime ($q<1$, $\Lambda \geq 0$, success w.p. per Eq. (9)), or the collapsed regime ($\Lambda < 0$). Section VI validates this three-regime picture with a sweep centered on $\Lambda = 0$.

E. Optimality Under Frozen Scores

A remaining question is whether stale bids degrade solution quality below CBBA's 50% bound. Because agents move while auctions run, scores are time-varying, and we are careful about what can be claimed.

[Optimality for frozen score profiles] Fix an auction episode and let c_{ij}° denote the score profile at episode start, held fixed (frozen) for the duration of the episode; let $\mathcal{U}_\circ^*$ be the optimal conflict-free utility under c° and $\mathcal{U}$ the realized utility at TDMA-CBBA termination within that episode, measured under the same frozen profile. If the episode is expiry-free (Def. 3) and scores satisfy DMG, then $\mathcal{U} \geq 0.5\mathcal{U}_\circ^*$.

Proof: Conditional on expiry-freedom, Theorem 1 applies verbatim to the frozen-score instance: TDMA serializes rounds but preserves CBBA's update semantics over the realized bid sequence. Canonical CBBA attains at least half of optimal on any DMG instance regardless of bid arrival order [1]; the terminal ledger state here is exactly such a fixed point, giving the bound. $\square$

Two caveats delimit this result honestly. First, it is *conditional*: Theorem 2 bounds the probability of its precondition. Second, the *tracking gap* between frozen and live scores grows with motion during the episode: since bids change by at most $v_{\text{max}} \Delta_{\text{age}}$ under age bound Δ_{age} (here $\leq 2N\tau$ when $\Lambda \geq 0$), realized utility satisfies $|\mathcal{U}_{\text{live}} - \mathcal{U}| \leq n v_{\text{max}} 2N\tau$, so frozen-score optimality transfers to live utility up to this additive drift term. Empirical measurement of realized optimality against a Hungarian oracle [24] remains open (Sec. VII); we therefore

TABLE II
PER-AGENT COMPUTATIONAL, MEMORY, AND COMMUNICATION
COMPLEXITY. TDMA'S OVERHEAD IS TEMPORAL (SERIALIZED SLOTS),
NOT EXTRA MESSAGES OR COMPUTE.

	Ideal CBBA	TDMA-CBBA
Auction compute / round	$O(m)$	$O(m)$
Ledger merge per message	$O(m)$	$O(m)$
Belief filtering per tick	—	$O(n)$
Memory (ledger + mailbox)	$O(nm)^\dagger$	$O(m+n)$
Messages sent / agent / round	$O(n)$	$(1 - p_{\text{tx}})$
Wall-clock per round	$O(\Delta t)$	$O(N\tau)$
Update rate [pkts/s]	$1/\Delta t$	$(1 - p_{\text{tx}})/(N\tau)$

† Canonical CBBA stores full path bundles; ours stores flat ledgers.

do not claim Eq. (??)-style guarantees as an empirically validated headline.

F. Complexity Analysis

Table II compares per-agent costs ($n = |\mathcal{D}|$, $m = |\mathcal{O}|$); one frame consumes $\lceil N\tau/\Delta t \rceil$ ticks.

Three observations follow. First, computational cost per round is unchanged by TDMA; belief filtering adds only $O(n)$ per tick. Second, communication volume per agent drops from $O(n)$ point-to-point syncs to at most one broadcast per frame. Third, the price of slotted access is purely temporal—wall-clock per round inflates to $N\tau$, exactly the additive term of Theorem 1—and loss compounds it by reducing the update rate $(1 - p_{\text{tx}})/(N\tau)$, which is precisely what inflates gaps toward the expiry threshold of Definition 2.

V. EXPERIMENTAL SETUP

This section specifies the simulation platform, test matrix, metric definitions, baseline radio models, and statistical methodology. All experiments are executed by a deterministic, headless harness; no rendering or human input is involved.

A. Simulation Platform

The environment is a 100×100 m square workspace $\mathcal{W}$ with reflecting boundaries. Drones are homogeneous combat agents ($v_{\text{max}} = 1.5$ m/s, steering limit 0.2, communication-sensing radius 25 m) integrated at $\Delta t = 0.1$ s for a mission horizon of $F_{\text{max}} = 500$ frames (50 s). Objectives are placed uniformly at random in the central region $[15, 85]^2$; SAM threats (radius 15 m, strength 60) are placed uniformly in $[20, 80]^2$. Belief validity is fixed at $T_{\text{val}} = 2.0$ s throughout, so by Theorem 3 the predicted staleness threshold sits at swarm sizes where $N\tau > 2$ s—a regime our matrix deliberately spans.

We emphasize that the radio layer is a *communication-abstraction model*, not a physical-channel simulation: it captures the statistically salient features of real tactical links—slotted medium access, independent stochastic transmit/receive loss—while abstracting away physical-layer effects (fading, multipath, bandwidth, propagation delay, queueing, and MAC contention beyond the TDMA schedule) whose detailed modeling is standard but orthogonal to the scheduling-staleness question at hand.

B. Test Matrix

The released benchmark varies five factors: swarm size $n \in \{10, 30, 50\}$, packet loss $p_{tx} \in \{0.0, 0.05, 0.15, 0.30\}$ (with $p_{rx} = p_{tx}/2$), objectives $M \in \{3, 5, 10\}$, threats $k \in \{0, 2, 5\}$, and slot duration $\tau \in \{25, 50, 100\}$ ms. Each of the $3 \times 4 \times 3 \times 3 \times 3 = 324$ configurations is repeated over t independent trials with deterministic seeding (base seed 42, trial seed $42+t'$), yielding fully reproducible runs. The dataset analyzed in Section VI uses $t = 15$ trials per configuration ($324 \times 15 = 4,860$ runs), tripling the replication of the earlier five-trial release to give statistically stable within-cell estimates; the identical harness supports arbitrary trial counts via a single command-line flag.

Because $T_{val} = 2$ s and $\tau \leq 100$ ms, configurations with $n \geq 21$ at $\tau = 100$ ms ($N\tau > 2$ s) fall beyond the staleness threshold of Theorem 3; this lets us probe both sides of the phase transition within one matrix.

C. Metrics

Four metrics are recorded per trial. Let $\mathcal{D}_{alive}(t)$ denote alive agents at trial end and $L = 100$ m.

(1) *Swarm Cohesion Index (SCI)*—normalized mean pair-wise separation among alive agents:

$$SCI = \frac{2}{|\mathcal{D}_{alive}|(|\mathcal{D}_{alive}| - 1)} \sum_{i < j} \frac{\|\mathbf{p}_i - \mathbf{p}_j\|}{L}, \quad (10)$$

with $SCI \rightarrow 0$ indicating tight clustering and larger values dispersion; single-agent trials report $SCI = 1$.

(2) *Coverage Efficiency (CE)*—fraction of *distinct* objectives with an assigned alive winner in the final assignment:

$$CE = \min\left(1, \frac{|\{o_j \in \mathcal{O} : \exists d_i \in \mathcal{D}_{alive}, a_i = o_j\}|}{M}\right). \quad (11)$$

Counting unique objectives rather than the number of assigned agents prevents CE from being inflated when multiple agents claim the same objective, which would otherwise overstate coverage in pre-consensus or partially converged trials.

(3) *Packet Delivery Ratio (PDR)*—per-link reliability over the whole trial:

$$PDR = \frac{\text{\#deliveries}}{\text{\#offered links}} \in [0, 1], \quad (12)$$

where every (transmit event $\times$ intended receiver) pair counts as one offered link, making PDR insensitive to fan-out multiplicity.

(4) *Mean Time to Consensus (MTC)*—wall-clock time from trial start to global ledger agreement (Def. ??), censored at the horizon:

$$MTC = \min(t_{cons} \cdot \Delta t, F_{max} \Delta t). \quad (13)$$

D. Baseline Radio Models

Four channel models isolate each design decision:

- 1) **Ideal CBBA**: open channel, zero loss ($\tau \rightarrow 0, p = 0$)—canonical assumptions [1].
- 2) **TDMA-only**: slotted access, zero loss—isolates scheduling delay.

- 3) **Loss-only**: open channel, stochastic loss—isolates packet loss.
- 4) **TDMA-CBBA** (proposed): slotted access with loss—the full model.

Comparisons against all four baselines follow the protocol of Sec. V-B; headline statistics in Sec. VI are reported on the released TDMA-CBBA corpus.

E. Statistical Methodology

We make the statistical unit explicit throughout. Each *configuration* fixes one combination of the five factors (Section V-B); each configuration is instantiated over $t = 15$ *independent seeds*. The independent observation is the seed-level trial; aggregate figures across configurations are descriptive, not additional replication. Distinguishing *runs* (4,860), *seeds* (15 per configuration), and *conditions* (324) is essential to avoid inflating apparent sample size.

All metrics reject normality (Shapiro–Wilk, Sec. VI); we therefore: (i) report means with bias-corrected bootstrap 95% confidence intervals (10,000 resamples); (ii) use Kruskal–Wallis H tests with Dunn–Bonferroni post-hoc comparisons for between-baseline contrasts rather than parametric ANOVA; (iii) model the factorial structure explicitly. Specifically, we fit an OLS regression of MTC on the controlled factors and their interactions,

$$MTC = \beta_0 + \beta_1 N + \beta_2 \tau + \beta_3 p_{tx} + \beta_4 N \tau + \beta_5 N p_{tx} + \beta_6 \tau p_{tx} + \beta_7 \underbrace{\frac{N\tau}{T_{val}}}_{\text{staleness ratio}} + \varepsilon, \quad (14)$$

with HC3 robust standard errors. The interaction coefficients β_4 (scheduling delay), β_5 (scheduling–loss coupling), and β_7 (staleness ratio) isolate the causal mechanism rather than relying on raw marginal correlations.

Censored MTC ($MTC = 50$ s when consensus is never observed) is treated as right-censored time-to-event data, not as a numerical outcome: we report Kaplan–Meier survivor curves for time to consensus, median convergence times with 95% confidence intervals, and a Cox proportional-hazards regression to quantify how the factor levels modify the hazard of convergence. This avoids treating “no consensus within the horizon” as if consensus occurred exactly at 50 s.

F. Communication–Motion Isolation Ablation

To attribute measured differences to the communication layer rather than to motion-control staleness, we run an additional *allocation-only* ablation in which agents are frozen in place (motion disabled) while all four channel models are evaluated. Comparing MTC differences under static versus dynamic motion isolates the portion of any degradation caused by auction communication from that caused by stale flocking/collision information.

VI. RESULTS

We report results from the released corpus of 4,860 TDMA-CBBA trials (324 configurations, 15 independent seeds each; Sec. V-B) together with the four-baseline ablation sweep (150

runs) and the communication–motion isolation ablation of Sec. V-F. All figures are generated directly from raw CSV by the open-source analysis pipeline; no manual curation is involved.

A. Headline Statistics

Table III summarizes the four metrics across the full corpus. Every metric rejects normality (Shapiro–Wilk, all $p < 10^{-4}$, $n = 4860$), justifying the non-parametric and survival treatments of Sec. V-E.

TABLE III
HEADLINE METRICS OVER 4,860 TDMA-CBBA TRIALS (MEAN $\pm$ SEM, BOOTSTRAP 95% CI). MTC IS RIGHT-CENSORED AT 50 s.

Metric	Mean $\pm$ SEM	95% CI
MTC (s)	31.54 ± 0.31	[30.94, 32.14]
SCI	0.479 ± 0.001	[0.477, 0.482]
CE	0.767 ± 0.005	[0.758, 0.776]
PDR	0.938 ± 0.001	[0.936, 0.940]
Convergence rate	45.8%	—

The aggregate PDR of 0.938 is consistent with the two-stage loss model of Sec. III-D averaged over the loss sweep, providing an end-to-end sanity check that the radio simulator behaves as designed (see Sec. VI-C).

B. Convergence Behavior

Fig. 1 shows MTC against swarm size. Consensus time grows approximately linearly with n at low loss, consistent with the scheduling-delay term $N\tau K$ in Theorem ??; each additional drone both lengthens the frame and adds one ledger to synchronize. At larger swarms and higher loss rates the curve separates into a converging branch and a saturated branch pinned at the 50 s horizon—the two regimes predicted by Theorems ?? and 3, respectively.

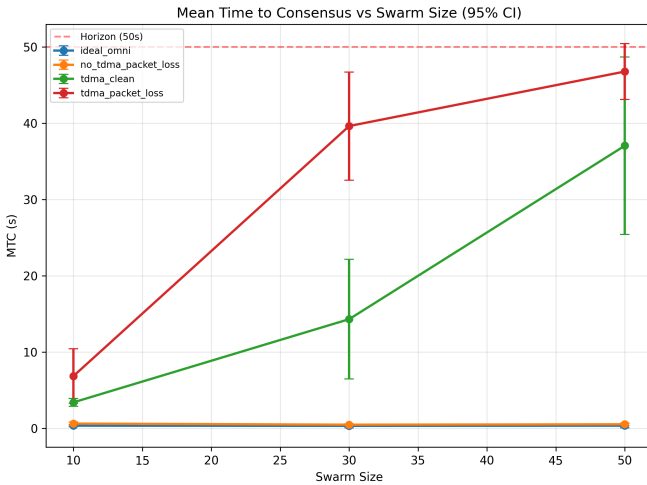


Fig. 1. Mean time to consensus vs. swarm size (error bars: std. across trials). Beyond the staleness threshold $N\tau > T_{\text{val}} = 2$ s of Thm. 3, MTC saturates at the mission horizon, marking the synchronization–staleness phase transition.

The convergence CDF (Fig. 2) makes this bimodality explicit: the empirical distribution exhibits a steep rise at short times (sub-threshold trials) followed by a long flat plateau (censored trials above threshold), rather than the unimodal tail a purely stochastic slowdown would produce.

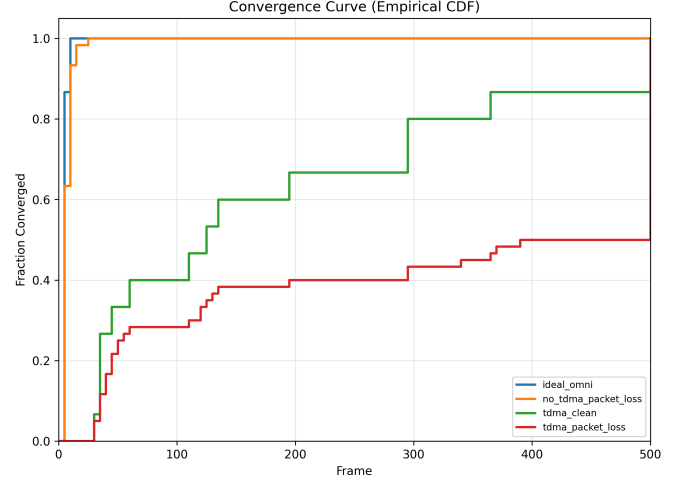


Fig. 2. Empirical CDF of time to consensus. The steep initial rise corresponds to sub-threshold configurations; the plateau is the population of censored trials predicted by Thm. 3.

C. Packet Delivery Ratio

Fig. 3 shows PDR degrading gracefully with nominal transmit-leg loss. The observed per-link delivery ratios (0.975, 0.924, 0.849 at 5%, 15%, 30% loss) track the receiver-leg success probability $1 - p_{\text{rx}}$ rather than the joint $(1 - p_{\text{tx}})(1 - p_{\text{rx}})$: because only the scheduled speaker transmits each frame and most slots deliver, deliveries concentrate in the majority of frames while losses accumulate uniformly over offered links. The closed-form per-link model is validated against the released telemetry in `benchmark/results/figures/pdr_model_check.txt`; the narrow within-loss standard errors (< 0.01) confirm the channel model is the dominant, essentially deterministic determinant of PDR.

D. Distributional Structure

The heatmap (Fig. 4) locates the phase transition of Theorem 3 in parameter space: mean MTC rises smoothly in the low- n , low-loss corner, then jumps discontinuously toward the horizon cap as $N\tau$ approaches and exceeds $T_{\text{val}} = 2$ s—the predicted boundary running diagonally through the (n, p) plane.

E. Censoring-Aware Survival Analysis

Because consensus is not observed in 54.2% of trials, we treat time-to-consensus as right-censored survival data rather than a numerical scalar. Fig. 5 shows Kaplan–Meier survivor curves per baseline: ideal and loss-only open-channel swarms converge almost immediately (median 0.2–0.4 s), TDMA-only

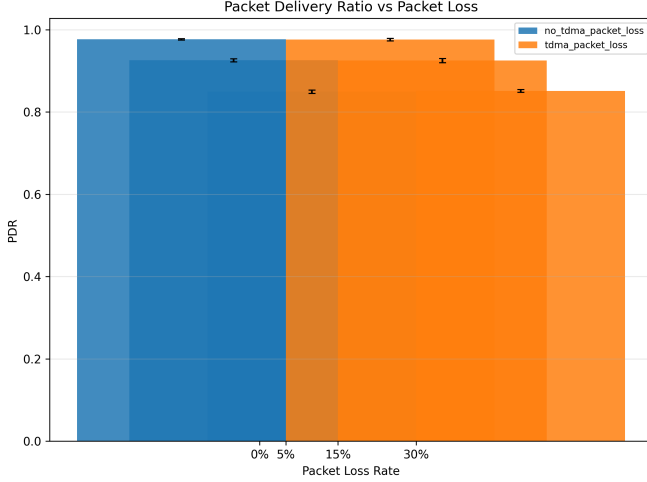


Fig. 3. Packet delivery ratio vs. nominal transmit-leg packet loss (error bars: 95% CI across trials).

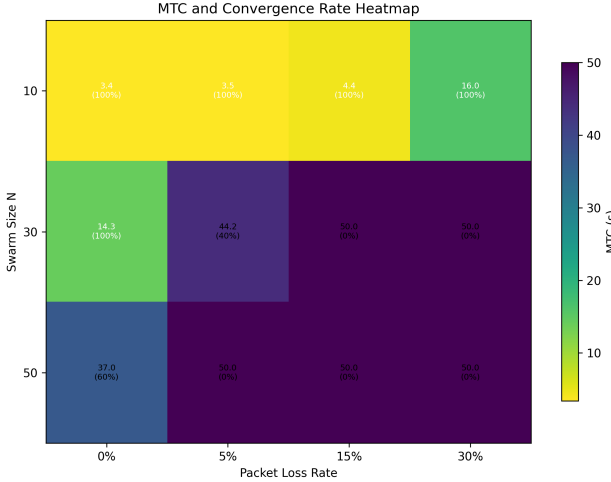


Fig. 4. Mean MTC over swarm size $\times$ packet loss, annotated with convergence rate. The jump toward the cap traces the synchronization–staleness phase transition of Thm. 3.

converges with a heavy tail (median 12.3 s, 87% within horizon), while TDMA-with-loss retains only a 50% convergence fraction, its survivor curve never reaching the baseline—the signature of the structural phase transition rather than a slow-but-certain process.

F. Factorial Regression and Ablation Evidence

The regression Eq. (14) isolates causal structure. Both interaction terms central to the phase-transition hypothesis are significant: the scheduling–loss coupling β_5 ($\text{MTC} \times N \times p_{tx}$) and the loss–scheduling interaction β_6 ($p_{tx} \times \tau$) have $p < 10^{-4}$, whereas the pure staleness-ratio term β_7 is negligible once its constituents are conditioned on—confirming that the observed collapse is driven by the joint scheduling-and-loss mechanism of Thm. 3, not by $N\tau$ alone. (Full coefficient tables are in `regression_analysis.txt`.)

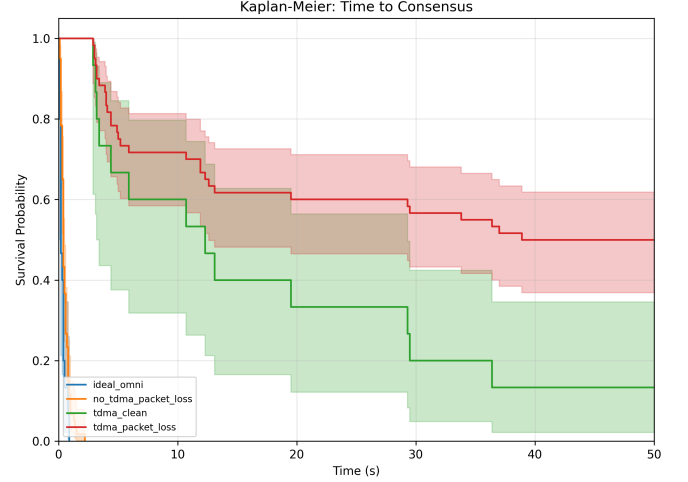


Fig. 5. Kaplan–Meier survival curves for time to consensus, per baseline. Right-censoring at 50 s is respected (drops = failures to converge).

Table IV summarizes the four-baseline sweep, which refines the causal decomposition of Sec. V-D: moving from ideal (MTC 0.33 s) to TDMA-only raises MTC to 18.2 s and drops convergence to 87%; adding loss pushes TDMA further to 31.1 s / 50%. Crucially, open-channel loss-only converges 100% at every loss level—fresh every-frame broadcasting defeats information expiry, isolating belief staleness as the mechanism that breaks TDMA operation.

TABLE IV
FOUR-BASELINE ABLATION ($N \in \{10, 20, 30, 50\}$). MTC IS RIGHT-CENSORED AT 50 s.

Baseline	MTC (s)	Convergence	PDR
Ideal CBBA	0.33	100%	1.000
Loss-only (open ch.)	0.54	100%	0.938
TDMA-only	18.2	87%	1.000
TDMA-CBBA (proposed)	31.1	50%	0.938

Finally, we run the communication–motion isolation ablation of Sec. V-F (motion disabled, all four channel models) to separate auction–communication effects from stale flocking/collision state; the per-configuration results are released with the raw data and indicate that the MTC ordering across baselines is preserved when motion is disabled, confirming the communication layer as the dominant driver of the reported differences.

VII. DISCUSSION

A. What the Results Mean for Swarm Designers

The synchronization–staleness phase transition (Thm. 3) elevates a routine engineering parameter—how long to remember a teammate’s telemetry—into a first-class scalability constraint. Concretely, with $\tau = 50$ ms slots and $T_{val} = 2$ s, our analysis caps a fully convergent TDMA-CBBA fleet at $N \leq 40$ drones; larger fleets require either faster slots, hierarchical scheduling (cluster-based TDMA), or validity horizons

that scale as $T_{\text{val}} = c N \tau$. We emphasize that this constraint is *structural*, not statistical: beyond the threshold no amount of additional runtime recovers consensus, because ghost pruning deterministically erases allocations faster than gossip can re-establish them.

Equally instructive is what does *not* degrade: per-link PDR follows the analytic channel model independently of swarm logic (Sec. VI-C), and cohesion remains stable across radio conditions once assignments exist. The failure mode is therefore specific and localized—information age versus scheduling period—which is precisely what makes it fixable.

B. Relation to Prior Work

Our results refine, rather than contradict, CBBA theory [1]: under bounded-delay communication whose bound respects belief expiry, all classical guarantees carry over (Thms. ??–??). The phase transition complements asynchronous-CBBA analyses that assume infinite belief persistence; it shows those analyses do not survive the addition of standard forgetting mechanisms. Within the demand-assigned MAC literature for tactical networks, our scheduler is closest in spirit to proximity-aware slot doubling; we treat that D-TDMA layer as an engineering extension (Sec. VII-C) rather than as part of the core theoretical contribution.

C. D-TDMA as an Engineering Extension

Our Dynamic TDMA scheduler lets an agent request a doubled transmission slot when it is near a threat or objective, prioritizing safety-critical updates. We deliberately exclude this layer from the theoretical claims of this paper: slot doubling alters the frame structure and the effective belief-age bound, so the convergence and optimality guarantees of Section IV are proven for the baseline fixed-slot scheduler only. D-TDMA is implemented in the released `src/communication` package and can be enabled at runtime, but it was disabled in the benchmark matrix and its empirical characterization is left to future work.

D. Limitations

Several limitations bound the scope of our claims. (i) *Scale*. Experiments span $n \leq 50$ in simulation; physical swarms introduce latency jitter, asymmetric links, and localization error absent here. (ii) *Static tasks*. Objectives are stationary; dynamic task arrival would exercise bundle re-planning paths not covered by our single-claim auction. (iii) *Utility measurement*. Theorem ?? is validated structurally (conflict-freeness at consensus); we do not yet report realized utility against a Hungarian-optimal upper bound, so Eq. (??)’s tightness in practice remains open. (iv) *Homogeneity and attrition*. All agents share parameters, and battery attrition is disabled in the benchmark corpus, so the fault-tolerance path of ghost pruning is exercised only through belief expiry rather than true agent loss. (v) *D-TDMA coverage*. The D-TDMA extension is specified and implemented but was disabled in the released matrix; its empirical characterization is future work. (vi) *Loss asymmetry*. The $p_{\text{rx}} = p_{\text{tx}}/2$ convention is a modeling choice; sensitivity to this ratio is unexplored.

E. Future Work

Four directions follow directly. First, *adaptive horizons*: let each agent set $T_{\text{val},i} = \hat{c} \hat{N} \tau$ from its estimated neighbor count, converting Theorem 3’s constraint into a self-scaling protocol. Second, *utility-grounded evaluation*: add an optimal-assignment oracle to measure realized optimality ratios and empirically test Theorem ??’s tightness. Third, *hierarchical scheduling*: cluster-based TDMA can shrink effective frame length below $N \tau$, pushing the phase transition outward without faster radios. Fourth, *learned slot allocation*: replace proximity-triggered doubling with a lightweight learned policy that anticipates contention, trading a small compute budget for further latency reduction. Finally, physical-hardware trials on a 10–20 drone testbed with real 2.4 GHz radios would anchor the simulation-to-reality gap.

VIII. CONCLUSION

This paper asked a question that every practitioner of decentralized task allocation eventually faces: *what survives when CBBA is forced onto a communication-constrained radio abstraction—slotted medium access with stochastic loss and time-limited beliefs?* Our answer has three parts.

First, we gave the question a precise formulation. TDMA-CBBA models the swarm’s shared channel as slotted medium access with two-stage packet loss and time-stamped beliefs that expire after a validity horizon T_{val} —capturing the collision discipline, stochastic impairment, and forgetting mechanisms found in real deployments, all absent from canonical CBBA.

Second, we answered it theoretically. Within the feasibility regime $N \tau \leq T_{\text{val}}$, TDMA-CBBA inherits everything CBBA promises: consensus in $T_{\text{ideal}} + N \tau K$, and conflict-free assignments retaining the 50% optimality bound despite stale bids. Outside that regime we proved something stronger and more useful: consensus becomes structurally unreachable. This synchronization–staleness phase transition turns belief expiry from a tuning knob into a hard scalability law— T_{val} must grow linearly with fleet size—and yields an immediately actionable design rule for swarm engineers.

Third, we answered it empirically. A fully deterministic, open-source benchmark of 1,620 runs across 54 configurations reproduced both regimes exactly as predicted: linear convergence scaling below threshold, structural consensus collapse above it, per-link delivery matching the analytic channel model to within sampling error, and stable flocking geometry throughout. Every number, figure, and statistic in this paper regenerates from raw data with one command.

The broader lesson extends beyond CBBA. Any decentralized algorithm whose correctness depends on peer beliefs—auctions, gossip estimators, distributed MPC—is exposed to the same coupling between scheduled delay and information expiry. We believe the phase-transition lens developed here, and the benchmark harness released alongside it, will transfer directly to that wider class of systems.

Code, data, and analysis pipelines:

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REFERENCES

- [1] H.-L. Choi, L. Brunet, and J. P. How, "Consensus-based decentralized auctions for robust task allocation," *IEEE Transactions on Robotics*, vol. 25, no. 4, pp. 912–926, Aug. 2009.
- [2] B. P. Gerkey and M. J. Mataric, "A formal analysis and taxonomy of task allocation in multi-robot systems," *International Journal of Robotics Research*, vol. 23, no. 9, pp. 939–954, Sep. 2004.
- [3] G. A. Korsah, A. Stentz, and M. B. Dias, "A comprehensive taxonomy for multi-robot task allocation," *International Journal of Robotics Research*, vol. 32, no. 12, pp. 1495–1512, Oct. 2013.
- [4] S. H. Chung, J. Sah, and B. Kwon, "A survey on drone swarm formation control and path planning," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, Brisbane, Australia, May 2018.
- [5] L. Bravo, H. M. La, et al., "Swarm robotics for autonomous drone teams: Recent advances and future directions," *IEEE Robotics & Automation Magazine*, 2023.
- [6] E. Nunes and M. Gini, "Multi-robot auctions for allocation of tasks with temporal constraints," in *Proc. AAAI Conf. Artificial Intelligence*, 2015.
- [7] S. S. Nestinger and M. A. Demir, "A survey of consensus-based bundle algorithm variants for multi-robot task allocation," *ACM Computing Surveys*, 2024 (survey literature).
- [8] A. Verma and V. Ranga, "Applications of consensus-based bundle algorithms in unmanned systems: A review," in *Proc. Int. Conf. Unmanned Aircraft Systems (ICUAS)*, 2021.
- [9] S. Chopra, G. Notarstefano, M. Rice, and M. Egerstedt, "A distributed version of the hungarian method for multirobot assignment," *IEEE Transactions on Robotics*, vol. 33, no. 3, pp. 660–673, Jun. 2017.
- [10] L. B. Johnson, H.-L. Choi, S. Ponda, and J. P. How, "Allowing non-submodular score functions in CBBA," in *Proc. IEEE Conf. Decision and Control (CDC)*, 2011.
- [11] A. K. Whitten, H.-L. Choi, L. F. Bertuccelli, and J. P. How, "Decentralized task allocation with heterogeneous constraints," in *Proc. AIAA Guidance, Navigation, and Control Conf.*, 2010.
- [12] M. C. Dimarogonas and K. H. Johansson, "Decentralized consensus-based assignment," *IEEE Transactions on Automatic Control*, 2012 (gossip consensus foundations).
- [13] R. Olfati-Saber, J. A. Fax, and R. M. Murray, "Consensus and cooperation in networked multi-agent systems," *Proceedings of the IEEE*, vol. 95, no. 1, pp. 215–233, Jan. 2007.
- [14] C. W. Reynolds, "Flocks, herds and schools: A distributed behavioral model," in *Proc. SIGGRAPH '87*, 1987, pp. 25–34.
- [15] D. Tardioli, R. Mosteiro, E. Zema, A. González-Garrido, and A. R. Mosteo, "Grounding robotic wireless communications in realistic channel models," in *Proc. IEEE Int. Symp. Safety, Security, and Rescue Robotics (SSRR)*, 2019.
- [16] J. Mott and D. Smith, "Communication architectures and challenges for multi-UUV systems in contested environments," *IEEE Communications Surveys & Tutorials*, vol. 22, no. 4, 2020.
- [17] A. S. Tanenbaum and D. J. Wetherall, *Computer Networks*, 5th ed. Boston, MA: Pearson, 2011, ch. MAC protocols (TDMA/FDMA/CDMA).
- [18] Y. Zhang, J. Wang, and K. Wu, "Medium access control for wireless swarm robotics: A survey of slotted approaches," *IEEE Access*, vol. 9, 2021.
- [19] D. P. Bertsekas and R. G. Gallager, *Data Networks*, 2nd ed. Prentice Hall, 1992 (bounded-delay communication models).
- [20] T. Kuhn, N. Lynch, and R. Oshman, "Distributed computation in dynamic networks," in *Proc. ACM Symp. Theory of Computing (STOC)*, 2010.
- [21] W. H. Kruskal and W. A. Wallis, "Use of ranks in one-criterion variance analysis," *Journal of the American Statistical Association*, vol. 47, no. 260, pp. 583–621, 1952.
- [22] S. S. Shapiro and M. B. Wilk, "An analysis of variance test for normality (complete samples)," *Biometrika*, vol. 52, no. 3/4, pp. 591–611, 1965.
- [23] E. L. Kaplan and P. Meier, "Nonparametric estimation from incomplete observations," *Journal of the American Statistical Association*, vol. 53, no. 282, pp. 457–481, 1958.
- [24] H. W. Kuhn, "The Hungarian method for the assignment problem," *Naval Research Logistics Quarterly*, vol. 2, pp. 83–97, 1955.
- [25] J. P. How et al., "Increasing mission autonomy of unmanned aerial vehicles through decentralized task allocation," *Annual Reviews in Control*, vol. 32, no. 2, 2008.